Question

The compounds or complex ions $\mathbf{B}, \mathbf{C}, \mathbf{D}$ all contain metal $\mathbf{A}$, and their mass fractions in $\mathbf{B}, \mathbf{C}, \mathbf{D}$ are $26.30\%, 31.60\%, 26.80\%$ respectively. All three have high symmetry. $\mathbf{B}$ is sensitive to oxygen and can be obtained by reacting chloride $\mathbf{E}$ of $\mathbf{A}$ with $\mathbf{F}$ in diethyl ether or benzene under a nitrogen atmosphere. $\mathbf{F}$ is a sodium salt of an acidic organic compound $\mathbf{G}$, which is aromatic and contains approximately 9.15% hydrogen. $\mathbf{C}$ is obtained by replacing two adjacent ligands in $\mathbf{B}$ with chlorine, and $\mathbf{C}$ can also be obtained by reacting $\mathbf{E}$ and $\mathbf{F}$ in dimethoxyethane. Another method for preparing $\mathbf{C}$ is to react the thallium salt of $\mathbf{G}$ with $\mathbf{E}$, followed by reduction with $\mathrm{SnCl_2}$. $\mathbf{D}$ is a bimetallic derivative cationic complex. It is known that $\mathbf{C}$ can react with sodium thiolate to generate $\mathbf{H}$, and $\mathbf{H}$ can then react with $\mathrm{NiCl_2}$ to obtain $\mathbf{D}$, which contains approximately 8.465% nickel.

Which of the following options regarding the unknown species $\mathbf{A} - \mathbf{H}$ is correct:

A. All other options are incorrect.

B. The valence electrons of element $\mathbf{A}$ possess paired electrons.

C. The central element of $\mathbf{B}$ satisfies the EAN rule.

D. The structure of $\mathbf{C}$ possesses a threefold rotational axis.

E. $\mathbf{D}$ contains a ligand with two chemical environments.

F. $\mathbf{G}$'s chemical formula contains three elements

Answer

Correct Answer: A

Detailed Explanation

The entry point for this question is $\mathbf{G}$, which has acidity, the negative ion has aromaticity, and has approximately 9.15% hydrogen.

Consider common acidic compounds containing benzene rings:

The mass fraction of hydrogen in benzoic acid is $6/(12 \times 6 + 16 \times 2 + 6) = 5.5\%$, consider reducing the number of heavy atoms;

The mass fraction of hydrogen in phenol is $6/(12 \times 6 + 16 + 6) = 6.4\%$, the number of heavy atoms also needs to be reduced; therefore, $\mathbf{G}$ most likely does not contain oxygen.

CHECKPOINT 0.5 PTS

$\mathbf{G}$ most likely does not contain oxygen

Common acidic compounds containing benzene rings are difficult to match, consider five-membered ring aromatic compounds:

The cyclopentadienyl anion has aromaticity, and the mass fraction of hydrogen in cyclopentadiene is $6/(12 \times 5 + 6) = 9.1\%$, which matches.

Other pyrrole/pyran species can also be searched for, but the mass fractions do not match. Therefore, $\mathbf{G}$ is cyclopentadiene C_5H_6, and $\mathbf{F}$ is sodium cyclopentadienide C_5H_5Na. C_5H_6 has only two elements, so option F is incorrect.

CHECKPOINT 2 PTS

G is cyclopentadiene C_5H_6

CHECKPOINT 2 PTS

F is sodium cyclopentadienide C_5H_5Na

B is obtained from the reaction of the chloride of element **A** with sodium cyclopentadienide, which can be basically judged as a ligand substitution reaction; the chemical formula of **B** can be written as $\mathbf{A}(C_5H_5)_n$; determine the value of n based on the mass fraction:

CHECKPOINT 1 PTS

The generation of **B** is a ligand substitution reaction, and the chemical formula can be written as $\mathbf{A}(C_5H_5)_n$

If n is 1, the relative atomic mass of **A** is $65/(1-0.2630)-65=23$, which may be Na, but since **F** is NaC_5H_5, this possibility is excluded.

CHECKPOINT 1 PTS

F is NaC_5H_5, excluding the possibility that **A** is Na

If n is 2, the relative atomic mass of **A** is $130/(1-0.2630)-130=46.4$, which is not a reasonable divalent metal.

If n is 3, the relative atomic mass of $\mathbf{A}$ is $195/(1-0.2630)-195=69.6$, which is Ga.

If n is 4, the relative atomic mass of $\mathbf{A}$ is $260/(1-0.2630)-260=92.8$, which is Nb.

If n is 5 or more, there is no reasonable metal.

Therefore, it can be deduced that the metal $\mathbf{A}$ may be Ga or Nb, and the corresponding $\mathbf{B}$ may be $\mathrm{Ga(C_5H_5)_3}$ and $\mathrm{Nb(C_5H_5)_4}$.

CHECKPOINT 2 PTS

Metal $\mathbf{A}$ may be Ga or Nb

Then focus on $\mathbf{C}$, $\mathbf{C}$ is $\mathbf{B}$ in which two adjacent ligands are substituted by chlorine, then for these two metals, the chemical formulas of $\mathbf{C}$ are $\mathrm{GaCl_2(C_5H_5)}$ and $\mathrm{NbCl_2(C_5H_5)_2}$ respectively, calculate the mass fraction of the metal respectively:

If $\mathbf{A}$ is Ga, calculate $69.7/(65+35.5\times2+69.7)=33.9\%$, which is inconsistent with the given 31.6%.

If $\mathbf{A}$ is Nb, calculate $92.9/(65\times2+35.5\times2+92.9)=31.6\%$, which is consistent with the given condition.

Therefore, it is finally determined that the metal $\mathbf{A}$ is Nb; then $\mathbf{B}$ is $\mathrm{Nb(C_5H_5)_4}$, and $\mathbf{C}$ is $\mathrm{NbCl_2(C_5H_5)_2}$. $\mathbf{E}$ is $\mathrm{NbCl_4}$.

CHECKPOINT 2 PTS

Determine that the metal $\mathbf{A}$ is Nb through the mass fraction of $\mathbf{C}$

CHECKPOINT 0.5 PTS

B is $Nb(C_5H_5)_4$

CHECKPOINT 0.5 PTS

C is $NbCl_2(C_5H_5)_2$

The ground state valence electron configuration of Nb is $4d^45s^1$, there are no paired electrons, so option B is incorrect; in $Nb(C_5H_5)_4$, Nb is +4 valent, there is a single electron, and the EAN rule cannot be achieved no matter how the four cyclopentadienyl groups are coordinated, so option C is incorrect; **C** is $NbCl_2(C_5H_5)_2$, the structure is similar to dichloromethane, and there is no threefold axis, so option D is incorrect.

CHECKPOINT 0.5 PTS

The ground state valence electron configuration of Nb is $4d^45s^1$, there are no paired electrons

CHECKPOINT 0.5 PTS

$Nb(C_5H_5)_4$ cannot reach the EAN rule

CHECKPOINT 0.5 PTS

The structure of $NbCl_2(C_5H_5)_2$ is similar to dichloromethane, and there is no threefold axis

$NbCl_2(C_5H_5)_2$ can react with sodium thiomethoxide to generate $\mathbf{H}$, obviously a ligand substitution reaction occurs, the thiomethoxide anion substitutes the chloride ion, and the generated $\mathbf{H}$ is $Nb(CH_3S)_2(C_5H_5)_2$.

CHECKPOINT 1 PTS

The thiomethoxide anion substitutes the chloride ion, and the generated $\mathbf{H}$ is $Nb(CH_3S)_2(C_5H_5)_2$

$Nb(CH_3S)_2(C_5H_5)_2$ reacts with $NiCl_2$ to obtain the bimetallic derivative cationic complex $\mathbf{D}$, it can also be guessed that it is a ligand substitution reaction, and the product should not contain chlorine;

CHECKPOINT 1 PTS

The reaction to generate $\mathbf{D}$ can also be guessed as a ligand substitution reaction, and the product should not contain chlorine

The content of nickel in $\mathbf{D}$ is about 8.465%, assuming it contains one nickel atom, the remaining relative molecular mass is exactly two molecules of $Nb(CH_3S)_2(C_5H_5)_2$, so it can be calculated that the chemical formula of $\mathbf{D}$ is $[NiNb_2(CH_3S)_4(C_5H_5)_4]^{2+}$.

CHECKPOINT 2 PTS

It can be calculated that the chemical formula of $\mathbf{D}$ is $[NiNb_2(CH_3S)_4(C_5H_5)_4]^{2+}$

In the structure of this cationic complex, the cyclopentadienyl group is coordinated with Nb, and according to symmetry, there is only one chemical environment; Ni is obviously tetracoordinate, coordinated with four thiomethoxide groups, the thiomethoxide group is a bridging ligand coordinated with Ni and Nb, and there is also only one chemical environment, so option E is incorrect.

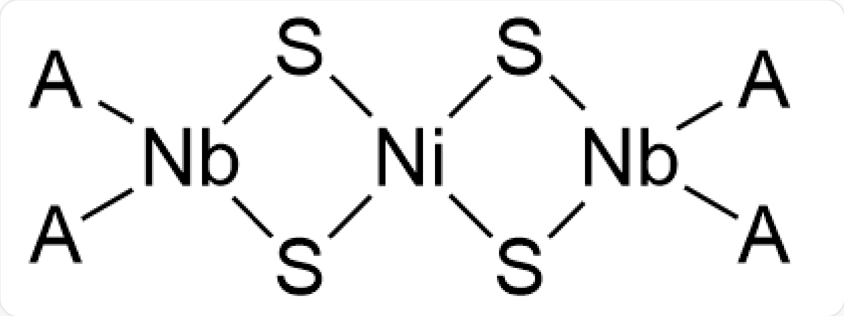
CHECKPOINT 1 PTS

The cyclopentadienyl group is coordinated with Nb, and according to symmetry, there is only one chemical environment

CHECKPOINT 1 PTS

The thiomethoxide group is a bridging ligand coordinated with Ni and Nb, and there is also only one chemical environment

The structure of the cationic complex is as follows:



D结构为：[A][Nb]1(S[Ni]2(S1)S[Nb]([A])(S2)[A])[A]；其中A代表环戊二烯基[cH-]1cccc1。

CHECKPOINT 1 PTS

D structure is: [A][Nb]1(S[Ni]2(S1)S[Nb]([A])(S2)[A])[A]; where A represents the cyclopentadienyl group [cH-]1cccc1

In summary, option A is correct.